

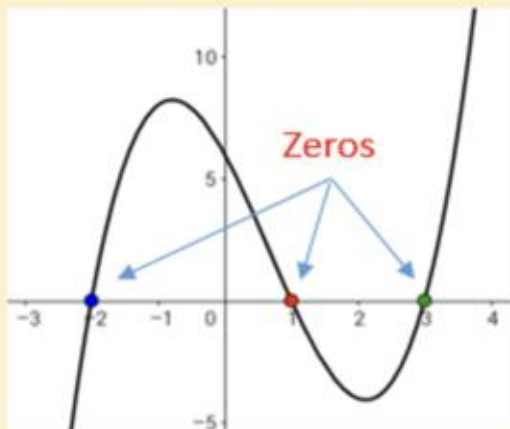
The Remainder & Factor Theorems

We know how to find solutions (zeros or roots) to linear (x^1) and quadratic (x^2) equations.

But what about higher order polynomials ?

The Zeros or Roots of a Polynomial Function

Graphically: the real zeros or real roots of a polynomial function are the x-intercepts of the graph.



Finding Roots and Factors of Polynomial Equations

The Factor Theorem

Assume we have a polynomial function of degree n

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

and c is a real number such that $P(c) = 0$.

1. The real number c is a zero or root of $P(x)$.
2. $x = c$ is a solution to the equation $p(x) = 0$.
3. $(x - c)$ is a factor of $P(x)$.

The **Factor theorem** says that if 2 is a zero of $P(x)$ then $(x - 2)$ is a factor of $P(x)$, and vice versa.

A simple **quadratic** function can be factorised by inspection e.g. Consider

$$f(x) = x^2 + 2x - 3$$

The factors are $(x-1)(x+3)$

From the factors, we can see that the **zeros** of the functions are

$$x = 1 \quad \text{and} \quad x = -3$$

As,

$$f(1) = (1)^2 + 2(1) - 3 = 0$$

$$f(-3) = (-3)^2 + 2(-3) - 3$$

$$= 9 - 6 - 3 = 0$$

Reversing this process enables us to factorise cubics (and polynomials of higher degrees)



Factoring Cubics

Consider a **cubic** function and first guessing possible zero values

e.g. If $f(x) = x^3 + 7x^2 + 9x + 3$

$$\begin{aligned}x = -1 \Rightarrow f(-1) &= (-1)^3 + 7(-1)^2 + 9(-1) + 3 \\&= -1 + 7 - 9 + 3 \\&= 0\end{aligned}$$

$$f(-1) = 0 \Rightarrow (x + 1) \text{ is a factor}$$

(Notice the change of sign)

For a polynomial function $f(x)$, the Factor Theorem says that:

If $f(a) = 0$, then $(x - a)$ is a factor

Factorising a **Cubic Function**

1. Factorise $x^3 - 2x^2 + 3x - 2$

Step 1 – find the first factor by trial & error (usually try values of $x = -1, 1, -2, 2, -3, 3$ etc)

Solution: $f(x) = x^3 - 2x^2 + 3x - 2$

Try $x = 1$: $f(1) = (1)^3 - 2(1)^2 + 3(1) - 2$
 $= 1 - 2 + 3 - 2 = 0$

$$f(1) = 0 \Rightarrow (x - 1) \text{ is a factor}$$

Once we have found one factor of a cubic, the other polynomial factor can be found by inspection.

Factorise

$$x^3 - 2x^2 + 5x + 8$$

Solution: We use the factor theorem to find one **linear** factor:

Let $f(x) = x^3 - 2x^2 + 5x + 8$

The linear factor can only contain numbers that are factors of 8.

We could try any of $x = \pm 1, \pm 2, \pm 4, \pm 8$

$$f(1) = 1 - 2 + 5 + 8 = 12 \neq 0 \Rightarrow (x - 1) \text{ is not a factor}$$

$$f(-1) = -1 - 2 - 5 + 8 = 0 \Rightarrow (x + 1) \text{ is a factor}$$

$$\text{So, } x^3 - 2x^2 + 5x + 8 = (x + 1) \times (\text{quadratic factor})$$

The quadratic factor can be found by inspection.


$$x^3 - 2x^2 + 5x + 8 = (x + 1)(x^2 + bx + 8)$$

We first get the $+x^3$ and $+8$

$$\therefore x^3 - 2x^2 + 5x + 8 = (x + 1)(x^2 + bx + 8)$$

To find the value of b , we look at the x^2 term and notice we must get $-2x^2$ after expansion

$$\therefore (x) \cdot (bx) + (1) \cdot (x^2) = -2x^2, \text{ so } b = -3$$

$$\therefore x^3 - 2x^2 + 3x - 2 = (x + 1)(x^2 - 3x + 8)$$

Note: $(x^2 - 3x + 8)$ does not factor further into whole number factors so we leave our answer in this form.

SUMMARY

Factorising Cubic Functions

- Use the factor theorem to find one linear factor

if $f(a) = 0$ then $(x - a)$ is a

factor

- Use inspection to find the quadratic factor
 - Start with the x^3 term of the cubic
 - Find the constant
 - Use the x^2 term of the cubic to find the middle term of the quadratic factor
 - Check factors using the x term of the cubic
- Factorise the quadratic factor if possible



Exercises



Factorise the following cubics:

1. $x^3 + 7x^2 + 14x + 8$

2. $x^3 + 7x^2 + 16x + 12$

3. $x^3 - 2x^2 + 2x - 4$

Exercises



Factorise the following cubics:

$$\begin{aligned} 1. \quad & x^3 + 7x^2 + 14x + 8 \\ &= (x + 1)(x^2 + 6x + 8) \\ &= (x + 1)(x + 2)(x + 4) \end{aligned}$$

$$\begin{aligned} 2. \quad & x^3 + 7x^2 + 16x + 12 \\ &= (x + 2)(x^2 + 5x + 6) \\ &= (x + 2)(x + 2)(x + 3) \end{aligned}$$

$$\begin{aligned} 3. \quad & x^3 - 2x^2 + 2x - 4 \\ &= (x - 2)(x^2 + 2) \end{aligned}$$



2 Find a given that:

a $(x - 1)$ is a factor of $2x^3 - 3x^2 + ax - 4$

3 Find the constant k and hence factorise the polynomial if:

a $2x^3 + x^2 + kx - 4$ has the factor $(x + 2)$

3 a $k = -8, P(x) = (x + 2)(2x + 1)(x - 2)$

Remainder Theorem

If a polynomial $P(x)$ is divided by a linear **expression** $(x - a)$ the result will be a polynomial quotient $Q(x)$ and a remainder $R(x) = P(a)$.

Evaluating Remainders

Example 24

Self Tutor

Use the Remainder theorem to find the remainder when $x^4 - 3x^3 + x - 4$ is divided by $x + 2$.

If $P(x) = x^4 - 3x^3 + x - 4$, then

$$\begin{aligned} P(-2) &= (-2)^4 - 3(-2)^3 + (-2) - 4 \\ &= 16 + 24 - 2 - 4 \\ &= 34 \end{aligned}$$

$\therefore$ when $x^4 - 3x^3 + x - 4$ is divided by $x + 2$, the remainder is 34. {Remainder theorem}

Ex: Find the remainder when $x^3 - x^2 + 2x + 4$ is divided by $2x + 1$

Solution: Let $f(x) = x^3 - x^2 + 2x + 4$

To find the value of a , we let $2x + 1 = 0$. The value of x gives the value of a .

$$2x + 1 = 0 \Rightarrow x = -\frac{1}{2} \quad \text{so,} \quad R = f\left(-\frac{1}{2}\right)$$

$$f\left(-\frac{1}{2}\right) = \left(-\frac{1}{2}\right)^3 - \left(-\frac{1}{2}\right)^2 + 2\left(-\frac{1}{2}\right) + 4$$

$$\Rightarrow f\left(-\frac{1}{2}\right) = -\frac{1}{8} - \frac{1}{4} - 1 + 4$$

$$\Rightarrow R = \frac{-1 - 2 - 8 + 32}{8} \Rightarrow R = \frac{21}{8}$$

1. Find the remainder when $x^3 + 4x^2 + 3x - 5$

is divided by $x + 1$

Exercises



1. Solution:

$$f(x) = x^3 + 4x^2 + 3x - 5$$

When dividing by $(x + 1)$, the remainder $= f(-1)$

$$\Rightarrow f(-1) = (-1)^3 + 4(-1)^2 + 3(-1) - 5$$

$$\Rightarrow R = -1 + 4 - 3 - 5$$

$$\Rightarrow R = -5$$

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2. If $x + 2$ is a factor of $x^3 + ax^2 + bx - 6$ and if the remainder on division by $x - 1$ is -3 , find the values of a and b .
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Exercises

2. Solution:



If $x + 2$ is a factor of $x^3 + ax^2 + bx - 6$ and if the remainder on division by $x - 1$ is -3 , find the values of a and b .

Solution: Let

$$f(x) = x^3 + ax^2 + bx - 6$$

$$f(-2) = 0 \Rightarrow (-2)^3 + a(-2)^2 + b(-2) - 6 = 0$$

$$\Rightarrow 4a - 2b - 14 = 0$$

$$\Rightarrow 2a - b = 7 \quad \text{--- (1)}$$

$$f(1) = -3 \Rightarrow 1 + a + b - 6 = -3$$

$$\Rightarrow a + b = 2 \quad \text{--- (2)}$$

$$(1) + (2) \Rightarrow 3a = 9 \Rightarrow a = 3$$

$$\text{Substitute in (2)} \Rightarrow b = -1$$



6. Use the remainder theorem to find the remainder when $P(x)$ is divided by $(x - c)$ for the given value of c .

(a) $P(x) = 2x^3 - 3x^2 + 4x - 7, c = 2$

(b) $P(x) = x^5 - 2x^4 + 3x^2 - 20x + 3, c = -1$

(c) $P(x) = 5x^4 + 30x^3 - 40x^2 + 36x + 14, c = -7$

(d) $P(x) = x^3 - x + 1, c = \frac{1}{4}$



6. (a) By the remainder theorem, the required remainder is equal $P(2)$

$$P(2) = 2(2)^3 - 3(2)^2 + 4(2) - 7 = 5$$

(b) By the remainder theorem, the required remainder is equal $P(-1)$

$$P(-1) = (-1)^5 - 2(-1)^4 + 3(-1)^2 - 20(-1) + 3 = 23$$

(c) By the remainder theorem, the required remainder is equal $P(-7)$

$$P(-7) = 5(-7)^4 + 30(-7)^3 - 40(-7)^2 + 36(-7) + 14 = -483$$

(d) By the remainder theorem, the required remainder is equal $P\left(\frac{1}{4}\right)$

$$P\left(\frac{1}{4}\right) = \left(\frac{1}{4}\right)^3 - \frac{1}{4} + 1 = \frac{49}{64}$$